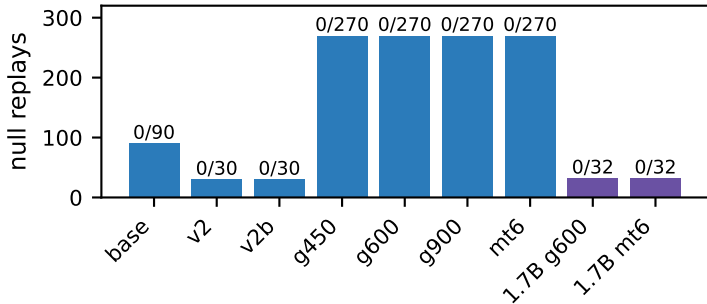


(a) APC off: every null replay exact ($|\Delta R| = 0$)



(b) prefix caching breaks determinism

